

# SIMATS ENGINEERING

## UNIT IV – ANALYTICAL ASSIGNMENT

COURSE CODE: ITA061

COURSE NAME: MACHINE LEARNING

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### QUESTION 1

Develop a K-Nearest Neighbour (KNN) model to classify soil fertility levels for improving crop productivity.

SDG Alignment: SDG 2 – Zero Hunger

#### 1. Problem Statement

The objective is to develop a K-Nearest Neighbour (KNN) model that classifies soil into fertility levels using soil-related parameters. The classification can help farmers understand soil conditions and support better crop productivity.

#### 2. Dataset Description

A publicly available agricultural or soil dataset can be used. Relevant attributes may include soil nutrient values such as nitrogen, phosphorus and potassium, along with pH and other available soil parameters. The target variable represents the soil fertility level.

#### 3. Data Preprocessing

The dataset should be checked for missing values and duplicate records. Missing numerical values can be handled using suitable imputation. Features should be normalized or standardized because KNN is distance-based. Irrelevant attributes should be removed before model training.

#### 4. KNN Model Development

KNN classifies a new soil sample by finding the K most similar samples in the training dataset. Similarity can be measured using Euclidean distance.

$$\text{Euclidean distance} = \sqrt{\sum (x_i - y_i)^2}$$

For example, if  $K = 5$ , the five nearest soil samples are selected. The fertility class occurring most frequently among these neighbours is assigned to the new soil sample. Different K values can be tested and the best value can be selected using validation.

#### 5. Model Evaluation

The model can be evaluated using Accuracy, Precision, Recall and F1-score. A confusion matrix can also be used to understand correct and incorrect classifications for each fertility category.

#### 6. Result Visualization

The results can be presented using a confusion matrix, class distribution chart, and suitable feature plots. These visualizations help interpret how well the KNN model separates different soil fertility levels.

## **7. Interpretation**

If the model achieves good evaluation scores, it indicates that the selected soil parameters contain useful information for distinguishing fertility levels. The predictions can support farmers in identifying soil conditions and making more informed crop-management decisions.

## **8. SDG-Based Recommendation**

This assignment is aligned with SDG 2 – Zero Hunger. A soil-fertility classification system can support sustainable agricultural decision-making by helping identify soil conditions and encouraging appropriate soil management practices to improve crop productivity.

## **9. Conclusion**

A KNN model can be developed to classify soil fertility levels using relevant soil parameters. Proper preprocessing, feature scaling, selection of K, and evaluation using classification metrics are important for obtaining reliable results. The solution has practical relevance to improving agricultural productivity and supports SDG 2 – Zero Hunger.

## **10. References**

1. Publicly available soil/agricultural dataset from a recognized repository such as Kaggle, UCI ML Repository, or Data.gov.in.
2. Machine Learning course materials – Unit IV: Instance-Based Learning.